

Probability Distribution

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#) Binomial Distribution:

A discrete random variable X is said to follow binomial distribution if the probability of getting ' r ' success is given by,

$$P(X=r) = {}^nC_r p^r q^{n-r} \quad (\forall r=0,1,2,\dots,n)$$

where,

n = number of trials

p = probability of success

$q = 1-p$ = probability of failure

Important points:

1) Binomial distribution is used to find probability of such event which has only two outcomes such as hit and miss, pass and fail, win and loss, head and tail, boy and girl etc.

2) The number of trials ' n ' must be fixed and finite.

3) The constant ' n ' and ' p ' used in binomial distribution are called its parameter.

4) For binomial distribution

$$\text{Mean} = np$$

$$\text{variance} = npq$$

$$\text{S.D} = \sqrt{npq}$$

5) Always suppose probability of success ' p ' to that event whose probability is to be determined

Fitting of Binomial Distribution:

Fitting of binomial distribution means finding of expected frequency (theoretical frequency) with the help of observed frequency (experimental frequency).

The expected frequency of getting 'r' success is,

$$f(r) = N P(X=r)$$

$$\therefore f(r) = N {}^n C_r p^r q^{n-r}$$

Notes

- 1) If the value of 'p' is not known then find the value of 'p' as below:

$$np = \bar{X}$$

$$\therefore p = \frac{\bar{X}}{n}$$

- 2) Total observed frequency and total expected frequency must be equal.

Poisson Distribution

A discrete random variable X is said to follow poisson distribution if the probability of happening of 'r' rare or casual event is given by,

$$P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$$

$$\forall r = 0, 1, 2, 3, \dots, \infty$$

Fitting of Poisson Distribution

Fitting of poisson distribution means finding of expected frequency (theoretical frequency) with the help of given observed frequency.

The expected frequency of getting 'r' rare of casual event is,

$$f(r) = N P(X=r)$$

$$\therefore f(r) = N \cdot \frac{e^{-\lambda} \lambda^r}{r!}$$

Note

- 1) If value of λ is not given then find the value of ' λ ' as,

$$\lambda = \bar{X} \text{ where, } \bar{X} = \frac{\sum fX}{N}$$

- 2) Total observed frequency and total expected frequency must be equal.

♥ Maaya I Love You very very much beyond infinity

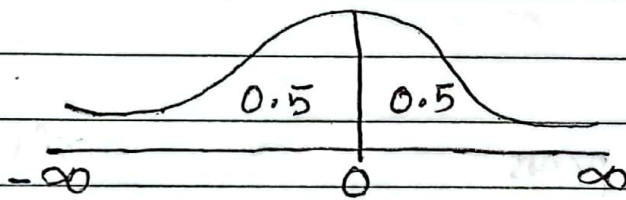
Normal Distribution

A continuous random variable X is said to follow normal distribution if its probability density function is given by,

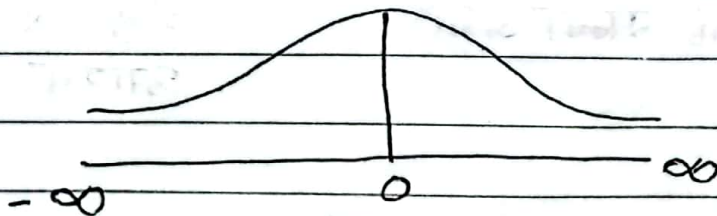
$$P(X) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

Note

- 1) $Z = \frac{X - \mu}{\sigma}$ is known as standard normal variable.
- 2) The constants μ and σ used in normal distribution are known as parameter of normal distribution.
- 3) For normal distribution:
 Mean = μ
 Standard deviation = σ
 Variance = σ^2
- 4) The normal curve is bell shaped and symmetrical.
- 5) The total area covered by normal curve is assumed to be '1' and it represents total probability.



- 6) The area given in normal table is for positive value of Z . So, whenever we have to find the area of negative value of Z then we must first make the value of Z positive by using the concept of symmetry.



$$P(0 < Z < 1.2) = 0.3849$$

$$P(0 < Z < 1.23) = 0.3907$$

$$P(-1.22 < Z < 0) = P(0 < Z < 1.22) = 0.3888$$

$$\# \quad 1.2 \Rightarrow \frac{1.2}{L} \quad \frac{.00}{R}$$

$$1.23 \Rightarrow \frac{1.2}{L} \quad \frac{.03}{R}$$

Note

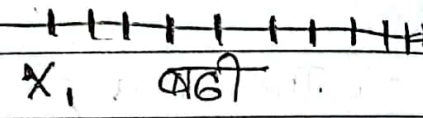
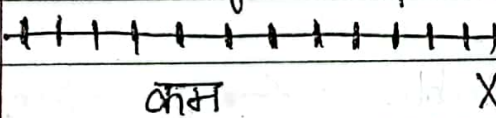
1) 10, 30, 37, 50, 51, 60, 70, 90

poor
fail
non-topper
young

Rich
pass to
topper
old

2) Highest
income of 20% poor

Lowest income of
richest 10%.



$$\rightarrow P(X < X_1) = 20\%$$

$$\rightarrow P(X > X_1) = 10\%$$

$\rightarrow$ shading left side बाट
सुकु गर्ने तर probability
अनुसार right सम्म जान
सकछ।

$\rightarrow$ shading right बाट
गर्ने तर probability
अनुसार left सम्म
आउन सकछ।

% दिया भने probability हो।

number दिया भने, Total % number गर्ने probability
निकालने